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0.1 Let's find the beautiful 2 direvative of this equation:

$$\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2$$

0.1.1 Let's find 1 direvative of the equation

AHAHAHAHAHHA

$$\begin{aligned} \frac{d}{dx} (\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2) \\ = \frac{d}{dx} (\sin(15 \cdot x^3 + 3)) + \frac{d}{dx} (\cos(5 \cdot x + 2)^2) \end{aligned}$$

AHAHAHAHAHHA

$$\frac{d}{dx} (\cos(5 \cdot x + 2)^2) = 2 \cdot (\cos(5 \cdot x + 2))^{2-1} \cdot \frac{d}{dx} (\cos(5 \cdot x + 2))$$

AHAHAHAHAHHA

$$\frac{d}{dx} (\cos(5 \cdot x + 2)) = 0 - \frac{d}{dx} ((5 \cdot x + 2)) \cdot \sin(5 \cdot x + 2)$$

AHAHAHAHAHHA

$$\frac{d}{dx} ((5 \cdot x + 2)) = \frac{d}{dx} (5 \cdot x) + \frac{d}{dx} (2)$$

AHAHAHAHAHHA

$$\frac{d}{dx} (2) = 0$$

Though

$$\frac{d}{dx} (2) = 0$$

AHAHAHAHAHHA

$$\frac{d}{dx} (5 \cdot x) = \frac{d}{dx} (5) \cdot x + 5 \cdot \frac{d}{dx} (x)$$

AHAHAHAHAHHA

$$\frac{d}{dx} (x) = 1$$

Obviously

$$\frac{d}{dx}(x) = 1$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(5) = 0$$

Though

$$\frac{d}{dx}(5) = 0$$

Obviously

$$\frac{d}{dx}(5 \cdot x) = (0 \cdot x + 5 \cdot 1)$$

Though

$$\frac{d}{dx}((5 \cdot x + 2)) = 0 \cdot x + 5 \cdot 1 + 0$$

Obviously

$$\frac{d}{dx}(\cos(5 \cdot x + 2)) = (0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2))$$

Though

$$\frac{d}{dx}(\cos(5 \cdot x + 2)^2) = \left(2 \cdot \cos(5 \cdot x + 2)^{(2-1)} \cdot (0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2))\right)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(\sin(15 \cdot x^3 + 3)) = \frac{d}{dx}((15 \cdot x^3 + 3)) \cdot \cos(15 \cdot x^3 + 3)$$

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$$\frac{d}{dx}((15 \cdot x^3 + 3)) = \frac{d}{dx}(15 \cdot x^3) + \frac{d}{dx}(3)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(3) = 0$$

Obviously

$$\frac{d}{dx}(3) = 0$$

АААААААААА

$$\frac{d}{dx}(15 \cdot x^3) = \frac{d}{dx}(15) \cdot x^3 + 15 \cdot \frac{d}{dx}(x^3)$$

АААААААААА

$$\frac{d}{dx}(x^3) = 3 \cdot (x)^{3-1} \cdot \frac{d}{dx}(x)$$

АААААААААА

$$\frac{d}{dx}(x) = 1$$

Though

$$\frac{d}{dx}(x) = 1$$

Obviously

$$\frac{d}{dx}(x^3) = (3 \cdot x^{(3-1)} \cdot 1)$$

АААААААААА

$$\frac{d}{dx}(15) = 0$$

Though

$$\frac{d}{dx}(15) = 0$$

Obviously

$$\frac{d}{dx}(15 \cdot x^3) = (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1)$$

Though

$$\frac{d}{dx}((15 \cdot x^3 + 3)) = 0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0$$

Obviously

$$\frac{d}{dx}(\sin(15 \cdot x^3 + 3)) = ((0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \cos(15 \cdot x^3 + 3))$$

Though

$$\begin{aligned} & \frac{d}{dx}(\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2) \\ &= (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \cos(15 \cdot x^3 + 3) + 2 \\ & \quad \cdot \cos(5 \cdot x + 2)^{(2-1)} \cdot (0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2)) \end{aligned}$$

0.1.2 Let's simplify this equation:

$$(0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \cos(15 \cdot x^3 + 3) + 2 \\ \cdot \cos(5 \cdot x + 2)^{(2-1)} \cdot (0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2))$$

Now it looks nice!

$$15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3) + 2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))$$

0.1.3 Let's find 2 direvative of the equation

АНАНАНАНАНАНА

$$\frac{d}{dx} (15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3) + 2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ = \frac{d}{dx} (15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3)) + \frac{d}{dx} (2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)))$$

АНАНАНАНАНАНА

$$\frac{d}{dx} (2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ = \frac{d}{dx} (2) \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) \\ + 2 \cdot \frac{d}{dx} (\cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)))$$

АНАНАНАНАНАНА

$$\frac{d}{dx} (\cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ = \frac{d}{dx} (\cos(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) \\ + \cos(5 \cdot x + 2) \cdot \frac{d}{dx} ((0 - 5 \cdot \sin(5 \cdot x + 2)))$$

АНАНАНАНАНАНА

$$\frac{d}{dx} ((0 - 5 \cdot \sin(5 \cdot x + 2))) = \frac{d}{dx} (0) - \frac{d}{dx} (5 \cdot \sin(5 \cdot x + 2))$$

АНАНАНАНАНАНА

$$\frac{d}{dx} (5 \cdot \sin(5 \cdot x + 2)) = \frac{d}{dx} (5) \cdot \sin(5 \cdot x + 2) + 5 \cdot \frac{d}{dx} (\sin(5 \cdot x + 2))$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} (\sin (5 \cdot x + 2)) = \frac{d}{dx} ((5 \cdot x + 2)) \cdot \cos (5 \cdot x + 2)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} ((5 \cdot x + 2)) = \frac{d}{dx} (5 \cdot x) + \frac{d}{dx} (2)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} (2) = 0$$

Obviously

$$\frac{d}{dx} (2) = 0$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} (5 \cdot x) = \frac{d}{dx} (5) \cdot x + 5 \cdot \frac{d}{dx} (x)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} (x) = 1$$

Though

$$\frac{d}{dx} (x) = 1$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx} (5) = 0$$

Obviously

$$\frac{d}{dx} (5) = 0$$

Though

$$\frac{d}{dx} (5 \cdot x) = (0 \cdot x + 5 \cdot 1)$$

Obviously

$$\frac{d}{dx} ((5 \cdot x + 2)) = 0 \cdot x + 5 \cdot 1 + 0$$

Though

$$\frac{d}{dx}(\sin(5 \cdot x + 2)) = ((0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2))$$

AHAHAHAHAHHA

$$\frac{d}{dx}(5) = 0$$

Obviously

$$\frac{d}{dx}(5) = 0$$

Though

$$\frac{d}{dx}(5 \cdot \sin(5 \cdot x + 2)) = (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2))$$

AHAHAHAHAHHA

$$\frac{d}{dx}(0) = 0$$

Obviously

$$\frac{d}{dx}(0) = 0$$

Though

$$\frac{d}{dx}((0 - 5 \cdot \sin(5 \cdot x + 2))) = 0 - (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2))$$

AHAHAHAHAHHA

$$\frac{d}{dx}(\cos(5 \cdot x + 2)) = 0 - \frac{d}{dx}((5 \cdot x + 2)) \cdot \sin(5 \cdot x + 2)$$

AHAHAHAHAHHA

$$\frac{d}{dx}((5 \cdot x + 2)) = \frac{d}{dx}(5 \cdot x) + \frac{d}{dx}(2)$$

AHAHAHAHAHHA

$$\frac{d}{dx}(2) = 0$$

Obviously

$$\frac{d}{dx}(2) = 0$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(5 \cdot x) = \frac{d}{dx}(5) \cdot x + 5 \cdot \frac{d}{dx}(x)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(x) = 1$$

Though

$$\frac{d}{dx}(x) = 1$$

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$$\frac{d}{dx}(5) = 0$$

Obviously

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Though

$$\frac{d}{dx}(5 \cdot x) = (0 \cdot x + 5 \cdot 1)$$

Obviously

$$\frac{d}{dx}((5 \cdot x + 2)) = 0 \cdot x + 5 \cdot 1 + 0$$

Though

$$\frac{d}{dx}(\cos(5 \cdot x + 2)) = (0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2))$$

Obviously

$$\begin{aligned} \frac{d}{dx}(\cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ = ((0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + \cos(5 \cdot x + 2) \\ \cdot (0 - (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2)))) \end{aligned}$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(2) = 0$$

Though

$$\frac{d}{dx}(2) = 0$$

Obviously

$$\begin{aligned} & \frac{d}{dx}(2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ &= (0 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + 2 \\ & \quad \cdot ((0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + \cos(5 \cdot x + 2) \cdot (0 - (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2)))) \end{aligned}$$

АНАНАНАНАНА

$$\begin{aligned} \frac{d}{dx}(15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3)) &= \frac{d}{dx}(15 \cdot 3 \cdot x^2) \cdot \cos(15 \cdot x^3 + 3) \\ &+ 15 \cdot 3 \cdot x^2 \cdot \frac{d}{dx}(\cos(15 \cdot x^3 + 3)) \end{aligned}$$

АНАНАНАНАНА

$$\frac{d}{dx}(\cos(15 \cdot x^3 + 3)) = 0 - \frac{d}{dx}((15 \cdot x^3 + 3)) \cdot \sin(15 \cdot x^3 + 3)$$

АНАНАНАНАНА

$$\frac{d}{dx}((15 \cdot x^3 + 3)) = \frac{d}{dx}(15 \cdot x^3) + \frac{d}{dx}(3)$$

АНАНАНАНАНА

$$\frac{d}{dx}(3) = 0$$

Though

$$\frac{d}{dx}(3) = 0$$

АНАНАНАНАНА

$$\frac{d}{dx}(15 \cdot x^3) = \frac{d}{dx}(15) \cdot x^3 + 15 \cdot \frac{d}{dx}(x^3)$$

АНАНАНАНАНА

$$\frac{d}{dx}(x^3) = 3 \cdot (x)^{3-1} \cdot \frac{d}{dx}(x)$$

АНАНАНАНАНА

$$\frac{d}{dx}(x) = 1$$

Obviously

$$\frac{d}{dx}(x) = 1$$

Though

$$\frac{d}{dx}(x^3) = (3 \cdot x^{(3-1)} \cdot 1)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(15) = 0$$

Obviously

$$\frac{d}{dx}(15) = 0$$

Though

$$\frac{d}{dx}(15 \cdot x^3) = (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1)$$

Obviously

$$\frac{d}{dx}((15 \cdot x^3 + 3)) = 0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0$$

Though

$$\frac{d}{dx}(\cos(15 \cdot x^3 + 3)) = (0 - (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \sin(15 \cdot x^3 + 3))$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(15 \cdot 3 \cdot x^2) = \frac{d}{dx}(15) \cdot 3 \cdot x^2 + 15 \cdot \frac{d}{dx}(3 \cdot x^2)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(3 \cdot x^2) = \frac{d}{dx}(3) \cdot x^2 + 3 \cdot \frac{d}{dx}(x^2)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(x^2) = 2 \cdot (x)^{2-1} \cdot \frac{d}{dx}(x)$$

АҺАҺАҺАҺАҺА

$$\frac{d}{dx}(x) = 1$$

Obviously

$$\frac{d}{dx}(x) = 1$$

Though

$$\frac{d}{dx}(x^2) = (2 \cdot x^{(2-1)} \cdot 1)$$

AHAHAHAHAHHA

$$\frac{d}{dx}(3) = 0$$

Obviously

$$\frac{d}{dx}(3) = 0$$

Though

$$\frac{d}{dx}(3 \cdot x^2) = (0 \cdot x^2 + 3 \cdot 2 \cdot x^{(2-1)} \cdot 1)$$

AHAHAHAHAHHA

$$\frac{d}{dx}(15) = 0$$

Obviously

$$\frac{d}{dx}(15) = 0$$

Though

$$\frac{d}{dx}(15 \cdot 3 \cdot x^2) = (0 \cdot 3 \cdot x^2 + 15 \cdot (0 \cdot x^2 + 3 \cdot 2 \cdot x^{(2-1)} \cdot 1))$$

Obviously

$$\begin{aligned} \frac{d}{dx}(15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3)) \\ = ((0 \cdot 3 \cdot x^2 + 15 \cdot (0 \cdot x^2 + 3 \cdot 2 \cdot x^{(2-1)} \cdot 1)) \cdot \cos(15 \cdot x^3 + 3) + 15 \cdot 3 \cdot x^2 \\ \cdot (0 - (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \sin(15 \cdot x^3 + 3))) \end{aligned}$$

Though

$$\begin{aligned} \frac{d}{dx}(15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3) + 2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))) \\ = (0 \cdot 3 \cdot x^2 + 15 \cdot (0 \cdot x^2 + 3 \cdot 2 \cdot x^{(2-1)} \cdot 1)) \cdot \cos(15 \cdot x^3 + 3) + 15 \cdot 3 \\ \cdot x^2 \cdot (0 - (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \sin(15 \cdot x^3 + 3)) \\ + 0 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + 2 \\ \cdot ((0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + \cos(5 \cdot x + 2) \\ \cdot (0 - (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2)))) \end{aligned}$$

0.1.4 Let's simplify this equation:

$$\begin{aligned} & (0 \cdot 3 \cdot x^2 + 15 \cdot (0 \cdot x^2 + 3 \cdot 2 \cdot x^{(2-1)} \cdot 1)) \cdot \cos(15 \cdot x^3 + 3) + 15 \\ & \cdot 3 \cdot x^2 \cdot (0 - (0 \cdot x^3 + 15 \cdot 3 \cdot x^{(3-1)} \cdot 1 + 0) \cdot \sin(15 \cdot x^3 + 3)) \\ & + 0 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) + 2 \\ & \cdot ((0 - (0 \cdot x + 5 \cdot 1 + 0) \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) \\ & + \cos(5 \cdot x + 2) \cdot (0 - (0 \cdot \sin(5 \cdot x + 2) + 5 \cdot (0 \cdot x + 5 \cdot 1 + 0) \cdot \cos(5 \cdot x + 2)))) \end{aligned}$$

Now it looks nice!

$$\begin{aligned} & 15 \cdot 3 \cdot 2 \cdot x \cdot \cos(15 \cdot x^3 + 3) + 15 \cdot 3 \cdot x^2 \cdot (0 - 15 \cdot 3 \cdot x^2 \cdot \sin(15 \cdot x^3 + 3)) \\ & + 2 \cdot ((0 - 5 \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) \\ & + \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot 5 \cdot \cos(5 \cdot x + 2))) \end{aligned}$$

0.2 Let's refresh our knowledge about the function

0.2.1 Function:

$$\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2$$

0.2.2 It's 2 derivative:

$$\begin{aligned} & 15 \cdot 3 \cdot 2 \cdot x \cdot \cos(15 \cdot x^3 + 3) + 15 \cdot 3 \cdot x^2 \cdot (0 - 15 \cdot 3 \cdot x^2 \cdot \sin(15 \cdot x^3 + 3)) \\ & + 2 \cdot ((0 - 5 \cdot \sin(5 \cdot x + 2)) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2)) \\ & + \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot 5 \cdot \cos(5 \cdot x + 2))) \end{aligned}$$

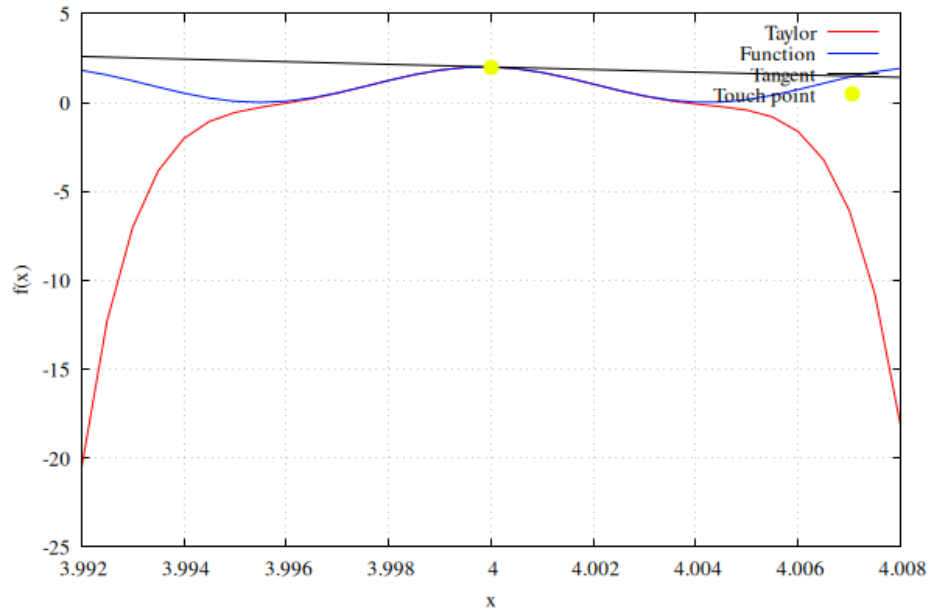
0.3 Let's research taylor's form of this equation:

$$\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2$$

0.3.1 This is taylor's form with first 7 summands around point 4

$$\begin{aligned} & 1.99 + -73.3 \cdot (x - 4) + -2.58e + 05 \cdot (x - 4)^2 + 6.2e \\ & + 06 \cdot (x - 4)^3 + 1.11e + 10 \cdot (x - 4)^4 + -1.53e + 11 \\ & \cdot (x - 4)^5 + -1.93e + 14 \cdot (x - 4)^6 + 1.73e + 15 \cdot (x - 4)^7 \end{aligned}$$

0.3.2 And this is a graph to compare func, it's tangent and taylor's form



0.4 Let's refresh our knowledge about the function

0.4.1 Function:

$$\sin(15 \cdot x^3 + 3) + \cos(5 \cdot x + 2)^2$$

0.4.2 It's first derivative:

$$15 \cdot 3 \cdot x^2 \cdot \cos(15 \cdot x^3 + 3) + 2 \cdot \cos(5 \cdot x + 2) \cdot (0 - 5 \cdot \sin(5 \cdot x + 2))$$